PROJECT REPORT OF SCIENCE EXHIBITION

BLINDSIGHT

AI-Powered Wearable Navigation Assistant for the Visually Impaired

PROJECT CREATORS

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I. NATURE OF WORK

✓ **Novel solution / Indigenous design to a problem / challenge**

II. RATIONALE

Visually impaired individuals often face difficulty navigating independently because they cannot easily identify obstacles or understand their surroundings. Existing assistive devices are often expensive or provide limited information.

The project 'Blindsight' is a low-cost wearable navigation assistant. An ESP32-CAM captures the environment and sends image data to a main ESP32 through ESP-NOW. The main ESP32 forwards it to a Raspberry Pi 3B+ local AI server and gives spoken guidance through earphones.

III. SCIENTIFIC PRINCIPLE(S) / CONCEPTS

The project is based on computer vision, local artificial intelligence, ESP-NOW and Wi-Fi wireless communication, ultrasonic ranging, digital signal processing, and text-to-speech synthesis.

The Raspberry Pi analyses the received image locally and produces a direction/summary. It converts this into WAV audio, which the main ESP32 sends through I2S to a PCM5102 DAC and earphones. Three HC-SR04 sensors run a fast rule-based fallback for nearby obstacles.

IV. MATERIALS USED

- ESP32 DevKit V1 (main controller)
- AI Thinker ESP32-CAM (OV2640)
- Raspberry Pi 3B+
- PCM5102 / PCM5100 DAC module
- 3 × HC-SR04 ultrasonic sensors
- Fallback push button
- 3.5 mm earphones / headphone amp
- 2 × Arduino Uno 5 V distribution rails
- Separate power banks / 5 V supplies
- ESP-NOW and Wi-Fi network
- Jumper wires and ECHO voltage dividers
- Headband, spectacle mount and 10×7×5 cm box
- Arduino IDE / PlatformIO
- Local Python AI and TTS server

V. PROCEDURE / DESCRIPTION

1. ESP32-CAM captures a compressed image of the environment.
2. Camera sends numbered image packets to the main ESP32 using ESP-NOW.
3. Main ESP32 reconstructs the image and says “Analysing”.
4. Main ESP32 sends the image to the Raspberry Pi 3B+ by local Wi-Fi.
5. Pi local AI analyses the scene and creates a safe direction/summary.
6. Pi converts the result to WAV speech and sends it back over Wi-Fi.
7. Main ESP32 plays WAV audio through I2S PCM5102 DAC and earphones.
8. On button press, three HC-SR04 sensors give immediate local obstacle guidance.

Future Scope: multilingual audio, GPS, improved local models, haptic alerts, and stronger obstacle detection.

VI. REFERENCES

- Raspberry Pi Documentation
- ESP32 / ESP-NOW Documentation
- ESP32-CAM OV2640 Documentation
- PCM5102 I2S DAC Documentation
- HC-SR04 Ultrasonic Sensor Documentation
- Arduino IDE / PlatformIO Documentation
- espeak-ng Text-to-Speech Documentation